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General and Comparative Endocrinology 135 (2004) 35–41
GENERAL AND COMPARATIVE ENDOCRINOLOGY
www.elsevier.com/locate/ygcen

The synthesis and effects of prostaglandins on the ovary of the crab *Oziotelphusa senex senex*

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Accepted 8 August 2003

Abstract

The possible involvement of prostaglandins in the regulation of ovarian development in the crab *Oziotelphusa senex senex* was investigated. Uptake of labelled arachidonic acid into the ovary of the crab was significantly greater than the other tissues. Prostaglandin H synthase activity was significantly increased in the ovary during the late vitellogenic stage when compared to immature ovary. The biosynthesis of different prostaglandins in the ovary was also measured during the crab reproductive cycle. Injection of prostaglandin $F_{2\alpha}$ and prostaglandin E_2 significantly increased ovarian index and oocyte diameter in a dose-dependent manner. In contrast, injection of prostaglandin D_2 did not affect ovarian growth. These results demonstrate the presence of prostaglandin biosynthetic system in ovary of the fresh water crabs.

Keywords: Oziotelphusa senex senex; Arachidonic acid; Prostaglandins; Prostaglandin H synthase; Ovarian maturation

1. Introduction

Prostaglandins (PGs) are the bioactive derivatives of C_{20} polyunsaturated fatty acids (PUFAs) widely distributed in mammalian tissues and implicated in diverse physiological functions. The first evidence for the presence of PGs in an invertebrate came from a study with the Caribbean gorgonian coral *Plexaura homomalla* (Weinheimer and Spraggins, 1969). In this species, Prostaglandin A_2 with potent hormone-like activity was found ten million times higher than those of other animals. PGA_2 induces severe vomiting in fish, and appears to be a defensive poison in *P. homomalla*. Since then, the occurrence and distribution of several PGs were reported in many invertebrates by different authors (see reviews by Stanley-Samuelson, 1994; Stanley-Samuelson and Pedibhotla, 1996). Even though the status of PG-research in invertebrates is fragmentary, there are several interesting reports pertaining to demonstration of these compounds and the diverse function they perform. In view of this an elaborate programme to identify and to study the physiological function of several PGs in invertebrates has been under taken in our laboratory. The present report is the first one in the series examines the uptake of arachidonic acid and identification of PGs in the tissues of fresh water edible crab *Oziotelphusa senex senex*. We have also determined the effect of $PGF_{2\alpha}$, PGD_2 and PGE_2 in the regulation of ovarian development in this crab.

2. Materials and methods

2.1. Animals

The crabs (*O. senex senex* Fabricius) were collected from paddy fields in and around Tirupati and maintained in the laboratory at 27 ± 1 °C in plastic tubs half submerged in tap water. They were acclimatized to laboratory conditions (12L:12D), for at least 10 days before being used in an experiment. Only intermolt female intact crabs with a body weight 30–32 g and carapace width 32–35 mm were used. The crabs were fed sheep meet daily ad libitum and the ambient medium

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doi:10.1016/j.ygcen.2003.08.002

was changed 6 h after feeding. Feeding was stopped 1 day before the commencement of the experiment to avoid variation in metabolism due to differential diet.

2.2. Chemicals

Arachidonic acid, PGD_2, PGE_2, and $PGF_{2\alpha}$ were purchased from Sigma. H^3 Arachidonic acid (100 Ci/mmol) is the generous gift from Prof. R. Ramamurthi, Department of Zoology, S.V. University, Tirupati, India. H^3 Arachidonic acid was prepared on request by Baba Atomic Research Centre, Mumbai, India. TLC plates of silica gel-60 (0.2 mm in thickness) were purchased from E. Merck, Germany.

2.3. Uptake of H^3-arachidonic acid in to different tissues of crab

The crabs were injected with 0.1 μCi H^3-arachidonic acid per 10 g body weight through the arthrodial membrane of the coxa of the fourth pair of walking legs with a hamilton micro-syringe. All the injections were given at 1000 h of the day. The crabs were sacrificed at different times (3, 6, and 12 h) after injection and the tissues were isolated for the estimation of arachidonic acid uptake. To the aliquots, 1 ml of diethyl ether and 5 ml of scintillation cocktail lipoluma was added and H^3 measurements were carried out by counting the samples on Rack-Beta liquid scintillation counter.

2.4. Assay of prostaglandin H synthase activity

Prostaglandin H synthase (EC 1.14.99.1) activity was measured in different tissues of the crab by the method of Padmavathi et al. (1990). The tissues after isolation were homogenized (20%, w/v) in 50 mM Tris–HCl, pH 8.0, containing 5 mM EDTA, and 5 mM diethyldithio carbamate (DDC). The homogenate was filtered through a cheese cloth and then centrifuged at 16,000*g* for 30 min in a refrigerated centrifuge. The supernatant was further centrifuged at 105,000*g* for 60 min at 4 °C. The resultant microsomal pellet was washed and rehomogenized in 50 mM Tris–HCl, pH 8.0, containing 0.1 M sodium perchlorate, 0.1 mM EDTA, and 0.1 mM DDC and recentrifuged for 60 min at 105,000*g*. The perchlorate washed microsomal pellet was subsequently homogenized in 50 mM Tris–HCl, pH 8.0, containing 0.1 mM EDTA and Tween 20 to the final concentration of 1%. The Tween 20 solubilized homogenate was centrifuged at 105,000*g* for 60 min at 4 °C and the clear supernatant was used as the source for prostaglandin H synthase.

The reaction mixture in a final volume of 3.0 ml contained 0.1 M Tris–HCl (pH 8.0), 5 mM L-Tryptophan, 2 mM EDTA, 1 μM Hematin, and the appropriate quantities of enzyme. Arachidonic acid was dissolved in small amount of 96% ethanol. The reaction was initiated by the addition of arachidonic acid to give a final concentration of 100 μM. The specific activity of the enzyme was expressed as micromoles of oxygen consumed/mg protein/min.

The protein content in the enzyme source was estimated by the method of Lowry et al. (1951) using bovine serum albumin as standard.

2.5. In vitro PG biosynthetic activity

We assessed PG biosynthesis in different tissues using the method developed by Padmavathi et al. (1990). The final reaction mixture containing 2 mg of protein, 10 μl of the co-factor mixture, and 0.2 μCi (H^3) arachidonic acid. The reaction volume was adjusted to 1.0 ml by adding buffer. After pre-incubation the reaction mixture for 10 min at 32 °C, the reaction was initiated by adding the microsomal preparation. After incubation at 32 °C for 2 min, the reaction was terminated by adding 176 μl of ethyl alcohol. After acidification to pH 4.0 with 0.1 M HCl, prostaglandins were extracted three times with 2.0 ml of ethyl acetate. The extracts were combined and evaporated under a stream of nitrogen. To assess spontaneous auto-oxidation of arachidonic acid, separate control tubes containing radio-labelled arachidonic acid along with boiled protein preparations were incubated and analyzed parallel with all experiments. The PG biosynthesis procedure was essentially similar to the one described earlier (Pedibhotla et al., 1997).

2.6. Isolation of prostaglandins in vivo

We also extracted PGs from the tissues in vivo. The protocol adopted in this study is essentially similar to the method published earlier (Padmavathi et al., 1990). In brief, the tissues were homogenized (10%, w/v) in 20% trichloro acetic acid. PGs were extracted three times by adding equal volume of ethyl acetate. The extracts were combined and evaporated under a stream of nitrogen. The reaction products ($PGF_{2\alpha}$, PGD_2, and PGE_2) were separated and identified by thin layer chromatography (TLC) and authenticated using the standards purchased from Sigma.

After evaporating the solvent, the extracted PGs were dissolved in acetone and spotted on TLC plates along with authentic standard PGs. The TLC plates were developed in a solvent system composed of chloroform:methanol:acetic acid:water (100:5:1:0.8). Individual PGs were made visible by spraying with 50% sulphuric acid followed by heating at 110 °C for 10 min. The products were visualized under UV light. Fig. 1 shows the chromatogram of standard PGs. Bands that corresponded in Rf to individual PGs were scrapped and then transferred to scintillation vials. Radio activity in each fraction was estimated by adding 1 ml of diethyl

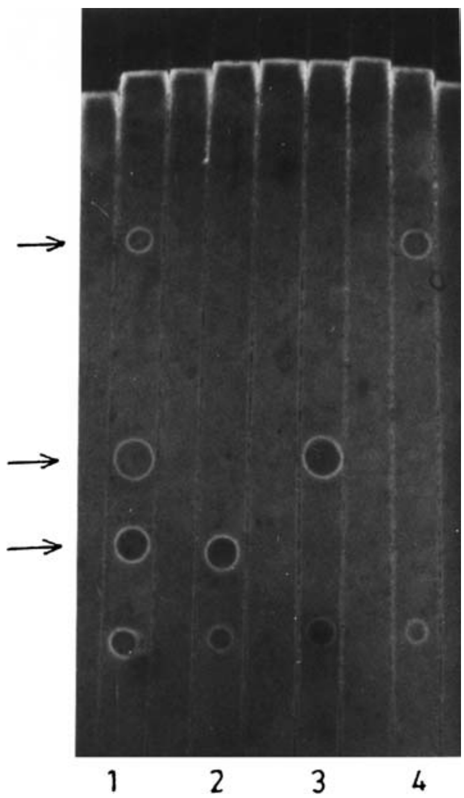

Fig. 1. Separation of prostaglandin standards by thin layer chromatography. Lane 1: mixture of $PGF_{2\alpha}$, PGE_2, and PGD_2; lane 2: $PGF_{2\alpha}$; lane 3: PGE_2; and lane 4: PGD_2. Arrows indicate the migration of prostaglandins.

ether and 5 ml of scintillation cocktail lipoluma and H^3 measurements were carried out by counting the samples on Rack-Beta liquid scintillation counter.

2.7. PG injection experimental protocol

The crabs were divided in to 11 groups of 20 animals each. First group served as initial control and this group of crabs, which received no treatment, was sacrificed on the first day of the experiment. Group 2 (concurrent control) received 10 μl of 5% ethanol in physiological saline (Van Harreveld, 1936) and groups 3, 4, and 5 injected with 10, 20, and 30 μg/crab $PGF_{2\alpha}$ respectively in 10 μl volume. Crabs in groups 6–8 were injected with 10, 20, and 30 μg/crab PGE_2 and crabs in groups 9–11 were received 10, 20, and 30 μg/crab PGD_2, respectively, in 10 μl volume. Stock solution in ethanol at 2 mg/ml was made up and PGs for injection was prepared as a suspension of 5% ethanol in crustacean saline. Injection in to groups 2–11 were administered on the 1st, 7th, and 14th day and sacrificed on 21st day.

After sacrifice, the crabs were weighed and the ovaries were removed, weighed and fixed in Bouin's fluid. After fixation for 24 h, the tissues were dehydrated in an alcoholic series, cleared in xylene, embedded in paraffin wax (m.p. 56–58 °C). Sections 6 μm thick were cut and stained with hematoxylin followed by counter staining with eosin (Bancroft and Stevens, 1982). The diameters of fifty oocytes in each ovary were measured using an ocular micrometer. Ovarian indexes were determined using the following formula.

$$\text{Ovarian index} = (\text{wet weight of ovary} / \text{wet weight of crab}) \times 100.$$

2.8. Statistical analysis

The data were analyzed using one way ANOVA followed by Duncan's multiple range test to determine the level of significance.

3. Results

3.1. Uptake of H^3 arachidonic acid into crab tissues

Uptake of arachidonic acid in different tissues at different time points was represented in Table 1. Among the tissues studied, uptake of labelled arachidonic acid was more in ovary when compared to other tissues. Data presented in the Table 2, show the uptake of H^3 arachidonic acid into the ovary during different

Table 1
Uptake of H^3-arachidonic acid into hepatopancreas, hypodermis, gill, and ovary of the crab, *Oziotelphusa senex senex* at 3, 6, and 12 h after isotope injection

Time (hours)	Hepatopancreas	Hypodermis	Gill	Ovary
3	8643 ± 967	9467 ± 314	6543 ± 468	18644 ± 814
6	12470 ± 1147	14783 ± 921	9437 ± 593	22967 ± 1256
	(44.28)	(56.15)	(44.20)	(23.19)
	<0.0001	<0.0001	<0.0001	<0.0001
12	16504 ± 1351	17431 ± 864	11634 ± 349	26479 ± 1031
	(90.95)	(84.00)	(77.81)	(42.02)
	<0.0001	<0.0001	<0.0001	<0.0001
F ratio	68.241	174.86	169.48	83.93
p value	<0.0001	<0.0001	<0.0001	<0.0001

Values are means (DPM/g wet weight of the tissue) ± SD of six individual crabs.
Values in the parentheses are % change from 3 h sample.

Table 2
Uptake of H^3-arachidonic acid into the ovary during different reproductive stages of the crab, *Oziotelphusa senex senex*

	Immature	Maturing	Fully matured
Colour of the ovary	White	Brown	Orange
Time post-injection (h)			
3	18644 ± 814	25437 ± 859	32197 ± 867
6	22967 ± 1256	31969 ± 1162	37732 ± 933
	(23.19)	(25.68)	(17.19)
	<0.0001	<0.0001	<0.0001
12	26479 ± 1031	35634 ± 751	49951 ± 1069
	(42.02)	(40.09)	(55.14)
	<0.0001	<0.0001	<0.0001
F ratio	83.93	181.08	537.24
p value	<0.0001	<0.0001	<0.0001

Values are means (DPM/g wet weight of the tissue) ± SD of six individual crabs.
Values in the parentheses are % change from 3 h sample.

reproductive stages of the crab. In general, the uptake of label increased with increase in time periods. The uptake of labelled arachidonic acid was significantly ($p < 0.0001$) more in late vitellogenic stage than in the immature stage of ovary.

3.2. Detection of prostaglandins in ovary and its quantification during different reproductive stages

The data from the Table 3 indicate that the major prostaglandin formed in the ovary was $PGF_{2\alpha}$ (60%), followed by PGE_2 (31%), and PGD_2 (9%). The amount of $PGF_{2\alpha}$ and PGE_2 formed was significantly higher in late vitellogenic stage of ovary when compared immature ovary. Where as the levels of PGD_2 were not altered significantly in the ovary during the reproductive cycle of the crab.

PG synthetic activity by isolated tissues was presented in Table 4. While the isolated mature ovaries yielded significantly more total PGs than the immature ovaries, the overall profile of individual PGs synthesized was similar in all the tissues.

3.3. Activity levels of prostaglandin H synthase

The distribution of prostaglandin H synthase in different tissues of the crabs is shown in Table 5. The prostaglandin H synthase activity levels were found to be higher in the ovary than somatic tissues. The prostaglandin H synthase activity was more during the late vitellogenic stage than the immature stage of ovary (Table 5).

3.4. Effect of PGs on the ovary

Fig. 2 shows the effects of graded concentrations of PGs on ovarian index of the crab. The changes in oocyte diameters after graded doses of PG administration is shown in Fig. 3. The mean ovarian index and oocyte diameter of the crabs that received $PGF_{2\alpha}$ were significantly ($p < 0.0001$) higher than the corresponding values for the saline injected concurrent control crabs (Figs. 2 and 3). With increasing doses of $PGF_{2\alpha}$ the ovarian index and oocyte diameters progressively increased. These results reveal not only that $PGF_{2\alpha}$ stimulated ovarian growth, but also that this stimulation was dose-dependent.

PGE_2 also stimulated ovarian growth only at higher concentrations tested (20 and 30 μg). The ovarian index and oocyte diameter of the crabs that received the smallest dose of PGE_2 did not differ significantly from the corresponding values of the concurrent control crabs (Figs. 2 and 3). With increasing doses of PGE_2, the ovarian index and oocyte diameters progressively increased. Where as injection of PGD_2 did not affect the ovarian growth in crabs (Figs 2 and 3).

Table 3
Levels of different prostaglandins produced by the ovary at different reproductive stages of crab, *Oziotelphusa senex senex*

Reproductive stage	$PGF_{2\alpha}$	PGE_2	PGD_2	*F* ratio	*p* value
Immature	3.61 ± 0.41	1.86 ± 0.12	0.52 ± 0.04	234.76	<0.0001
Maturing	5.22 ± 0.63	2.32 ± 0.18	0.54 ± 0.06	232.02	<0.0001
	(44.59)	(24.73)	(3.84)		
	<0.0004	<0.0004	NS		
Fully matured	9.34 ± 1.02	3.26 ± 0.26	0.53 ± 0.07	328.97	<0.0001
	(158.72)	(75.26)	(1.92)		
	<0.0001	<0.0001	NS		

Values are means (nmol/mg protein/h) ± SD of six individual crabs.
Values in the parentheses are % change from immature ovary. NS, not significant.

Table 4
In vitro PG biosynthesis by isolated tissues of the crab, *Oziotelphusa senex senex*

Tissue	PG $F_{2\alpha}$	PG E_2	PG D_2	F ratio	p value
Immature ovary	0.25 ± 0.032	0.098 ± 0.003	0.092 ± 0.001	65.973	<0.001
Maturing ovary	0.36 ± 0.026	0.14 ± 0.003	0.11 ± 0.002	432.098	<0.0001
	(44.00)	(42.85)	(19.56)		
	<0.0001	<0.0001	<0.0001		
Fully matured ovary	0.59 ± 0.043	0.22 ± 0.003	0.19 ± 0.002	341.548	<0.0001
	(136.00)	(124.48)	(106.52)		
	<0.0001	<0.0001	<0.0001		
Hepatopancreas	0.13 ± 0.015	0.062 ± 0.003	0.056 ± 0.002	242.651	<0.0001
	(−48.00)	(−36.73)	(−39.13)		
	<0.0001	<0.0001	<0.0001		
Hypodermis	0.16 ± 0.012	0.056 ± 0.002	0.054 ± 0.001	86.792	<0.0001
	(−36.00)	(−42.85)	(−41.30)		
	<0.0001	<0.0001	<0.0001		
Gill	0.09 ± 0.014	0.054 ± 0.002	0.047 ± 0.001	287.342	<0.0001
	(−64.00)	(−44.89)	(−48.91)		
	<0.0001	<0.0001	<0.0001		

Values are mean (pmol/mg protein/h) ± SD of six individual crabs.
Values in the parentheses are % change from immature ovary.

Table 5
The activity levels of prostaglandin H synthase in the tissues of the crab, *Oziotelphusa senex senex*

Tissue	Prostaglandin H synthase activity (μmol of O_2 consumed/mg protein/min.)
Immature ovary	1.23 ± 0.14
Maturing ovary	1.81 ± 0.19
	(47.15)
	<0.0001
Fully matured ovary	2.68 ± 0.24
	(117.8)
	<0.0001
Hepatopancreas	0.69 ± 0.08
	(−43.90)
	<0.0001
Gill	0.42 ± 0.06
	(−65.85)
	<0.0001
Hypodermis	0.56 ± 0.04
	(−54.47)
	<0.0001

Values are means ± SD of four individual crabs.
Values in the parentheses are % change from immature ovary.

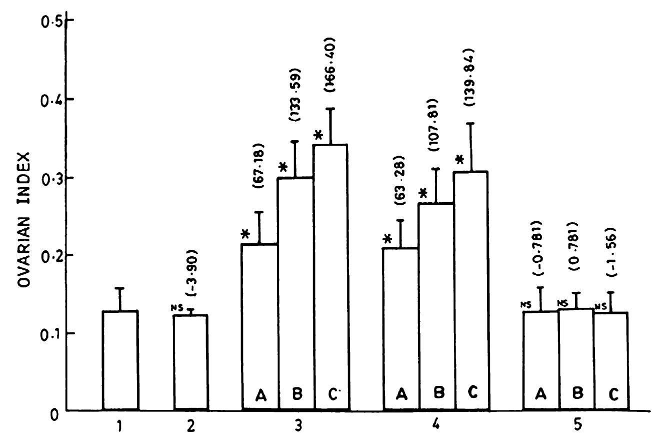

Fig. 2. Effect of graded doses ($A = 10$; $B = 20$; and $C = 30$ μg/crab) of $PGF_{2\alpha}$, PGE_2, and PGD_2 on the ovarian index of the crab, *Oziotelphusa senex senex*. 1: Initial control; 2: concurrent control; 3: $PGF_{2\alpha}$; 4: PGE_2; and 5: PGD_2. Values are means ± SD of 20 individual crabs. Values in the parentheses are % change from control crabs. $^*p < 0.0001$; NS, not significant.

4. Discussion

In decapod crustaceans, ovarian development and maturation are regulated by at least two neurohormones, the gonad inhibiting hormone from the X-organ-sinus gland complex in the eyestalks (Bomirsky et al., 1981; Otsu, 1963), and the gonad-stimulating hormone from the brain and thoracic ganglia (Eastman-Reks and Fingerman, 1984; Gomez, 1965). In addition to the two neurohormones, there is a non-neural hormone, methyl farnesoate that also controls the ovarian growth and maturation in crayfish *Procambarus clarkii* (Laufer et al., 1998) and crab *O. senex senex* (Reddy and Ramamurthi, 1998). Ovarian development in crustaceans can also be induced with injections of 5-hydroxytryptamine (Kulkarni et al., 1992), which acts indirectly on the ovaries, by inducing the release of gonad stimulating hormone.

In the present study the uptake of arachidonic acid into the ovary tissue increased during vitellogenesis. PG biosynthesis by isolated ovaries also increased during vitellogenesis. The activity of prostaglandin H synthase in the ovary increased in later stages of reproduction (vitellogenesis II). Considering the above and the fact that prostaglandins ($PGF_{2\alpha}$, PGE_2, and PGD_2) were demonstrated in the tissues of the crabs and more concentration in ovary, it seems likely that the ovarian growth in the crab is under the control of PGs.

Although there is a large body of information is available on the biological function of prostaglandins in vertebrates, relatively few data have been provided

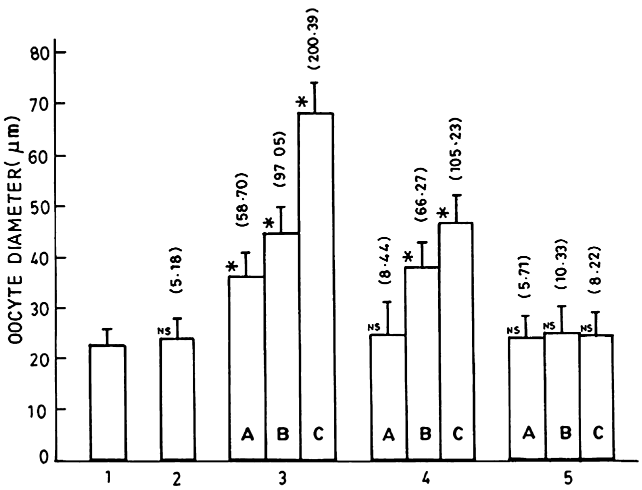

Fig. 3. Effect of graded doses ($A = 10$; $B = 20$; and $C = 30\,\mu$g/crab) of $PGF_{2\alpha}$, PGE_2, and PGD_2 on the oocyte diameter of the crab, *Oziotelphusa senex senex*. 1: Initial control; 2: concurrent control; 3: $PGF_{2\alpha}$; 4: PGE_2; and 5: PGD_2. Values are means ± SD of 20 individual crabs. Values in the parentheses are % change from control crabs. $^{*}p < 0.0001$; NS, not significant.

concerning their physiological activities in invertebrates. The regulatory role of prostaglandins in reproductive physiology is well established in mammals (Armstrong, 1981; Armstrong et al., 1974; Espey, 1980; Pride and Killick, 1993; Slater et al., 1994). The possible roles of eicosanoids in insects have been reviewed by Stanley-Samuelson and Pedibhotla (1996). Deridovich and Reunova (1993) summarized the participation of prostaglandins in the regulation of reproduction in bivalve molluscs. It was demonstrated earlier that PGE_2 stimulates a long-term increase in egg production in the snail *Helisoma durgi*, (Kunigelis and Saleuddin, 1986) and modulates the rate of proliferation of spermatogenesis in the scallop *Mizuhopecten yessoensis* (Varaskin and Reunova, 1993).

In crustaceans, the reproductive functions like releasing of hatching factors (Holland et al., 1985) and ovarian maturation (Sarojini et al., 1989) are known to be under the control of prostaglandins. Clare et al. (1985) showed that the barnacle hatching factor is a prostaglandin that co-migrated in thin layer chromatography with coral prostaglandin. Recently, Spaziani et al. (1993) reported the presence and gradual increase in ovarian PGE_2 and PGF_2 during vitellogenesis in the crayfish *Procambarus paeninsulanus*. The occurrence and effect of PGD_2, $PGF_{2\alpha}$, and PGE_2 in ovarian tissue and maturation was studied in the crayfishes *P. paeninsulanus* (Spaziani and Hinsch, 1997) and *Cherax quadricarinatus* (Silkovsky et al., 1998) and in the prawn *Macrobrachium rosenbergii* (Sagi et al., 1995). Further studies by Sagi et al. (1995) demonstrated that PGE_2 significantly induces cAMP synthesis in ovarian tissue in the prawn *M. rosenbergii*. Spaziani et al. (1993) observed that the rate of synthesis and the concentrations of $PGF_{2\alpha}$ and PGE_2 in the ovaries of the crayfish *P. paeninsulanus* increased during the final stages of yolk production.

The present work provides strong evidence that PGs are indeed involved in the regulation of reproduction in crustaceans in addition to gonad inhibiting hormone, gonad stimulating hormone, and methyl farnesoate. Further research must clarify the mechanisms of intermediation between the classical crustacean reproductive hormones (gonad inhibiting hormone, gonad stimulating hormone, and methyl farnesoate) and prostaglandins. Elucidation of action of PGs in crustaceans will be the basis for our future studies. Studies are also under way in this laboratory to test the effectiveness of PUFAs supplementing in diet in inducing ovarian development and spawning in brood stock females of commercially important crustaceans.

Acknowledgments

Special thanks to Prof. R. Ramamurthi, Sri Venkateswara University, Tirupati for providing H^3 Arachidonic acid and encouragement. We also thank Dr. K.V.S. Sarma, Associate Professor at S.V. University for statistical analysis of data. Mr. S. Umasankar provided valuable technical assistance. This work was generously supported by the Department of Science and Technology (SP/SO/CO4/96) and Indian Council of Agricultural Research (F. No. 4(65)/97/ASR-I) grants to PSR.

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